Fiscal Year:	FY 2012	**Task Last Updated:**	FY 05/25/2012
PI Name:	Mudawar, Issam Ph.D.		
Project Title:	Ground-based measurement and mechanistic modeling of dryout in reduced-gravity flow boiling, and transitioning to tests on the International Space Station (ISS)		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	47907-1288	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Fluid Physics NNH08ZTT002N
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No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	5	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX09AJ51A		
Performance Goal No.:			
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	Several efforts are presently underway to reduce the size and weight of space systems by transitioning from single-phase to two-phase thermal management. Because of large density differences between liquid and vapor, buoyancy can play an important role in defining the motion of vapor relative to liquid and, therefore, influence heat transfer effectiveness. This is particularly the case for future manned missions back to the Moon and afterwards to Mars. The emphasis in two-phase fluid physics is evident from recent NASA workshops that culminated in critical recommendations concerning Rankine cycle power conversion, thermal control systems and advanced life support systems. Among others, key fluid physics mechanisms that have been identified are flow boiling, dryout and condensation. A key strategy is to develop mechanistic models for flow conditions that would ensure insensitivity to gravity level for all these important heat transfer mechanisms. This would allow performing most of the required confirmatory experiments on Earth. Those		

Task Description:	three topics and the minimum flow requirements constitute key objectives for the proposed study. Unfortunately, most of the two-phase heat transfer know-how amassed over nearly a century of research comes from experiments that were conducted in Earth's gravity. The proposed study aims to develop an experimentally validated, mechanistic model for the minimum flow criteria to ensure gravity independent flow boiling heat transfer, dryout and condensation. Preparations for future intermediate parabolic flight experiments and long-term ISS experiments will occur throughout the course of the proposed study. This project will be a joint effort between the Purdue University Boiling and Two-Phase Flow Laboratory (BTPFL) and the NASA Glenn Research Center. Personnel from the two organizations combine extensive experience in research and development of both flow boiling and condensation systems, and in conducting microgravity experiments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	Previous models and correlations for the prediction of pressure drop in adiabatic and condensing mini/micro-channel flows have been validated for only a few working fluids and relatively narrow ranges of relevant parameters. A universal predictive approach for these flows must be capable of tackling many fluids with drastically different thermophysical properties and very broad ranges of all geometrical and flow parameters of practical interest. To achieve this goal, a new consolidated database of 7,115 frictional pressure gradient data points for both adiabatic and condensing mini/micro-channel flows is amassed from 36 sources. The database consists of 17 working fluids, hydraulic diameters from 0.0695 to 6.22 mm, mass velocities from 4.0 to 8,528 kg/m2s, liquid-only Reynolds numbers from 3.9 to 89,798, flow qualities from 0 to 1, and reduced pressures from 0.0052 to 0.91. It is shown that, while a few prior models and correlations provide fair predictions of the consolidated database, their predictive accuracy is highly compromised for certain subsets of the database. A universal approach to predicting two-phase frictional pressure drop is proposed by incorporating appropriate dimensionless relations in a separated flow model to account for both small channel size and different combinations of liquid and vapor states. This approach is shown to provide excellent predictions of the entire consolidated database and fairly uniform accuracy against all parameters of the database. This approach is also capable of tackling single and multiple channels as well as situations involving significant flow deceleration due to condensation. A new universal approach to predicting the condensation heat transfer coefficient for mini/micro-channel flows is proposed that is capable of tackling many fluids with drastically different thermophysical properties and very broad ranges of all geometrical and flow parameters of practical interest. This is accomplished by first amassing a consolidated database consisting of 4,045 data points from 28 sources. The database consists of single-channel and multi-channel data, 17 different working fluids, hydraulic diameters from 0.424 to 6.22 mm, mass velocities from 53 to 1,403 kg/m2s, liquid-only Reynolds numbers from 276 to 89,798, qualities from 0 to 1, and reduced pressures from 0.04 to 0.91. An exhaustive assessment of prior correlations shows only two correlations, that are actually intended for macro-channels, provide relatively fair predictions, while mini/micro-channel correlations generally show poor predictions. Two new correlations are proposed, one for predominantly annular flows, and the second for slug and bubbly flows. This approach shows very good predictions of the entire consolidated database, with an overall MAE of 16.0%. It is shown this accuracy is fairly even for different working fluids, and over broad ranges of hydraulic diameter, mass velocity, quality and pressure, and for both single and multiple mini/micro-channels. This report also explores the mechanism of flow boiling critical heat flux (CHF) for FC-72 in a 2.5 mm x 5 mm vertical upflow channel that is heated along its 2.5 mm sidewall downstream of an adiabatic development section. Unlike most prior CHF studies, where the working fluid enters the channel in liquid state, the present study concerns saturated inlet conditions with finite vapor void. Temperature measurements and high-speed video imaging techniques are used to investigate the influence of the inlet vapor void on interfacial behavior at heat fluxes up to CHF as well during the CHF transient. The flow entering the heated portion of the channel consists of a thin liquid layer covering the entire perimeter surrounding a large central vapor core. Just prior to CHF, a fairly continuous wavy vapor layer begins to develop between the liquid layer covering the heated wall and the heated wall itself, resulting in a complex four-layer flow consisting of the liquid layer covering the insulated walls, the central vapor core, the now separated liquid layer adjacent to the heated wall, and the newly formed wavy vapor layer along the heated wall. This behavior in captured in a new separated control-volume-based model that facilities the determination of axial variations of thicknesses and mean velocities of the four layers. Incorporating the results of this model in a modified form of the Interfacial Lift-off CHF Model is shown to provide fairly good predictions of CHF data for mass velocities between 185 and 1600 kg/m2s, evidenced by a mean absolute error of 24.52%.

Bibliography Type:	Description: (Last Updated: 01/06/2026)
Articles in Peer-reviewed Journals	Kharangate CR, Mudawar I, Hasan MH. "Experimental and theoretical study of critical heat flux in vertical upflow with inlet vapor void." International Journal of Heat and Mass Transfer, 2012 Jan;55(1-3):360-74. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2011.09.028 , Jan-2012
Articles in Peer-reviewed Journals	Kharangate CR, Mudawar I, Hasan MH. "Photographic study and modeling of critical heat flux in horizontal flow boiling with inlet vapor void." International Journal of Heat and Mass Transfer, 2012 Jul;55(15-16):4154-68. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2012.03.057 , Jul-2012
Articles in Peer-reviewed Journals	Kim SM, Mudawar I. "Universal approach to predicting two-phase frictional pressure drop for adiabatic and condensing mini/micro-channel flows." International Journal of Heat and Mass Transfer, 2012 May;55(11-12):3246-61. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2012.02.047 , May-2012
Articles in Peer-reviewed Journals	Kim SM, Mudawar I. "Consolidated method to predicting pressure drop and heat transfer coefficient for both subcooled and saturated flow boiling in micro-channel heat sinks." International Journal of Heat and Mass Transfer, 2012 Jun;55(13-14):3720-31. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2012.02.061 , Jun-2012
Awards	Mudawar I. "Certificate of Recognition in testimony of distinguished achievement of research in thermal engineering and service to engineering profession, The Japanese Society of Mechanical Engineers (JSME) and the American Society of Mechanical Engineers (ASME), March 2011." Mar-2011
Awards	Mudawar I. "Keynote Speaker: 'Two-Phase Micro-Channel Heat Sinks: Theory, Applications and Limitations,' ASME/JSME 2011 8th Thermal Engineering Joint Conference, Honolulu, Hawaii, March 2011." Mar-2011

Awards	Mudawar I. "Certificate of Recognition for 25 years of sustained contributions to the advancement of the arts, sciences and technology of aeronautics and astronautics, American Institute of Aeronautics and Astronautics (AIAA), January 2012." Jan-2012
Awards	Mudawar I. "Certificate of Recognition for 25 years of Service, American Society of Mechanical Engineers (ASME), March 2011." Mar-2011